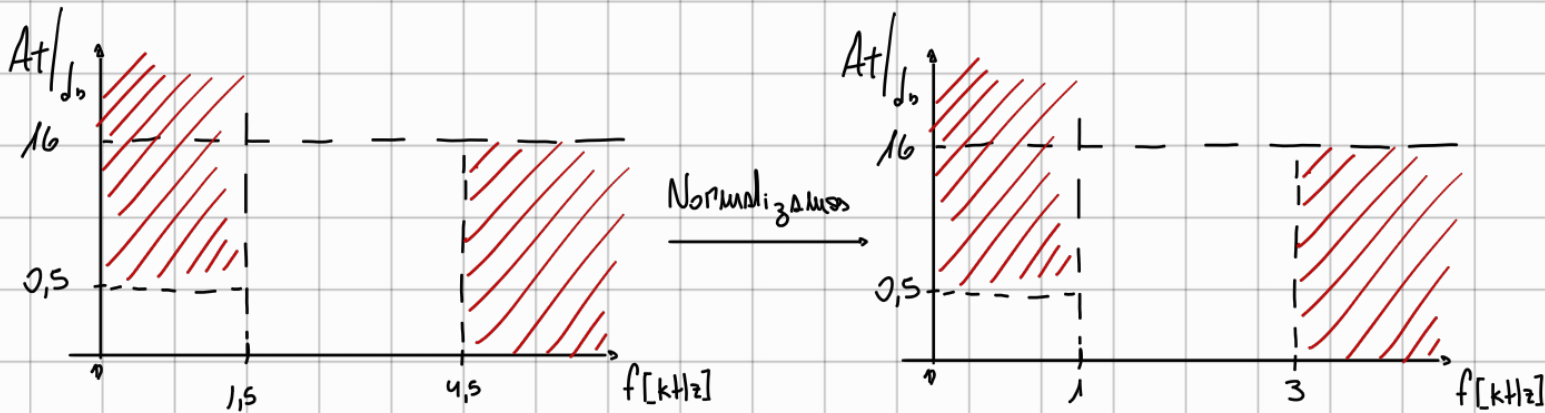


Trabajo práctico de laboratorio 1

Filtro	Función de aproximación	Frecuencia de corte	Frecuencia de stop	Atenuación máxima en banda de paso	Atenuación mínima en banda de stop
A	Chebyshev	1.5 kHz	4.5 kHz	0.5 dB	16 dB



$$\epsilon^2 = 10^{0.5/10} - 1 = 0.122$$

$$\alpha_n = 10 \log(1 + \epsilon^2 \cosh^2(n \cosh^{-1}(w_s)))$$

$$\alpha_n = 10 \log(1 + 0.122 \cosh^2(n \cosh^{-1}(3)))$$

$$n=2: 15.59 \text{ dB} < 16 \text{ dB} \quad \text{X}$$

$$n=3: 30 \text{ dB} > 16 \text{ dB} \quad \text{✓}$$

$$C_2(w) = 2w^2 - 1$$

$$C_3(w) = 4w^3 - 3w$$

$$H(jw)H(-jw) = \frac{1}{1 + \epsilon^2 C_3(w)^2} = \frac{1}{1 + \epsilon^2 (4w^3 - 3w)^2} = \frac{1}{1 + \epsilon^2 (16w^6 - 24w^4 + 9w^2)}$$

$$H(jw)H(-jw) = \frac{1/\epsilon^2}{16w^6 - 24w^4 + 9w^2 + 1/\epsilon^2} = \frac{1/\epsilon^2}{-16s^6 - 24s^4 - 9s^2 + 1/\epsilon^2}$$

$$H(jw)H(-jw) = \frac{1/16\epsilon^2}{-s^6 - \frac{3}{2}s^4 - \frac{9}{16}s^2 + \frac{1}{16\epsilon^2}} = \frac{1}{(s^3 + as^2 + bs + c)(s^3 + as^2 + bs + c)}$$

$$= \frac{1}{-s^6 - \frac{3}{2}s^4 - \frac{9}{16}s^2 + \frac{1}{16\epsilon^2}} = \frac{1}{-s^6 + (a^2 - 16)s^4 + (2ac - b^2)s^2 + c^2}$$

Igualamos término a término:

$$a = 1.15 \quad b = 1.53 \quad c = 0.715$$

$$H(s) = \frac{0,715}{s^3 + (1,253)s^2 + (1,535)s + 0,715}$$

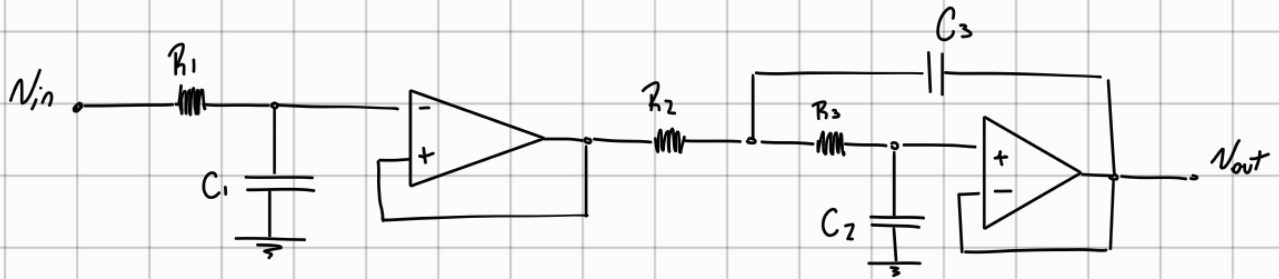
Factorización

para $s = -0,6258$

-0,6258	1	1,253	1,535	0,715
		-0,625	-0,39	-0,15
	1	0,627	1,142	ϕ

$$H(s) = \frac{0,715}{(s + 0,626)(s^2 + (0,627)s + 1,142)}$$

Circuito:



RC por Bajas:

$$H_1(s) = \frac{1}{sCR + 1} = \frac{1}{CR} \frac{1}{s + 1/CR}$$

$$H_1(s) = (0,626) \frac{1}{s + 0,626}$$

Sallen Key:

$$H_2(s) = \frac{G_2 G_3}{C_2 C_3} \frac{1}{s^2 + s \frac{1}{C_3} (G_2 + G_3) + \frac{G_2 G_3}{C_2 C_3}}$$

$$H_2(s) = (1,142) \frac{1}{s^2 + (0,627)s + 1,142}$$

$$R_1 = 1$$

$$G_2 = G_3 = 1$$

$$C_1 = \frac{1}{R_1 \cdot 0,625}$$

$$C_2 = \frac{G_1 G_2}{C_3 (1,142)}$$

$$0,627 = \frac{G_1 + G_2}{C_3} \Rightarrow C_3 = \frac{2}{0,627}$$

$$C_1 = 1,6$$

$$C_2 = 0,274$$

$$C_3 = 3,19$$

Desnormalización

$$C_1 = \frac{1,6}{(2\pi)(1,5\text{kHz})(R_N)} \rightarrow R_N = \frac{1,6}{(2\pi)(1,5\text{kHz})(10\text{nF})} \Rightarrow R_N = 1697\Omega$$

$$C_2 = \frac{0,274}{(2\pi)(1,5\text{kHz})(1697\Omega)} \rightarrow C_2 = 17\text{nF} \approx 15\text{nF}$$

$$C_3 = \frac{3,19}{(2\pi)(1,5\text{kHz})(1697\Omega)} \rightarrow C_3 = 200\text{nF}$$

$$R_1 = R_2 = R_3 = 1697\Omega$$

$$C_1 = 100\text{nF} \quad R_1 = 1697\Omega$$

$$C_2 = 15\text{nF} \quad R_2 = 1697\Omega$$

$$C_3 = 200\text{nF} \quad R_3 = 1697\Omega$$